



Strobogrammatic Number

A strobogrammatic number is a number that looks the same when viewed upside down or rotated 180 degrees. In other words, it is a number that remains unchanged when rotated by 180 degrees. The digits 0, 1, 8 are strobogrammatic digits, as they appear the same when viewed upside down or rotated. The digits 2 and 5 can also be considered strobogrammatic when they are rotated 180 degrees. However, the digits 3, 4, 6, and 9 are not strobogrammatic because they appear different when viewed upside down or rotated.

Strobogrammatic numbers are often used in number puzzles and games, and they can also be found in some real-world applications, such as odometers, digital clocks, and certain medical equipment.

Here are some examples of strobogrammatic numbers:

1. 69: This number looks the same when rotated 180 degrees. If you rotate it 180 degrees, you'll see that it still looks like 69.
2. 88: This number is also strobogrammatic. If you rotate it 180 degrees, it still looks like the same number.
3. 818: This number is a strobogrammatic number as well. If you rotate it 180 degrees, you'll see that it still looks like 818.
4. 101: This number is also strobogrammatic. If you rotate it 180 degrees, you'll see that it still looks like 101.
5. 906: This number is a strobogrammatic number because if you rotate it 180 degrees, you'll see that it looks like 609.

Note that in each of these examples, the digits that are strobogrammatic are 0, 1, and 8.

```
1. import java.util.HashMap;
2. import java.util.Map;
3. import java.util.Scanner;
4. public class Strobo {
5.     static boolean isStrobogrammatic(String num) {
6.         Map<Character, Character> map = new HashMap<Character, Character>();
7.         map.put('6', '9');
8.         map.put('9', '6');
9.         map.put('0', '0');
10.        map.put('1', '1');
11.        map.put('8', '8');
12.        int l = 0, r = num.length() - 1;
13.        while (l <= r) {
14.            if (!map.containsKey(num.charAt(l))) return false;
15.            if (map.get(num.charAt(l)) != num.charAt(r))
16.                return false;
17.            l++;
18.            r--;
19.        }
20.        return true;}
21.
```

```
22. public static void main(String args[])
23. {
24.     Scanner sc =new Scanner(System.in);
25.     System.out.print("Give num :");
26.     String n= sc.next();
27.     System.out.println(isStrobogrammatic(n));
28.
29. }
30. }
```



THANK YOU